

Proposed Solution

Date	03 October 2026
Team ID	17
Project Name	Fitbuddy-AI Fitness plan Generator
Department	Computer Technology
Institution / University	Madras Institute Of Technology / Anna University
Academic Year / Semester / Section	3 / 5 / Batch A
Faculty Mentor / Guide	Jayachitra V P
Team Leader	Bharath Jeyakkumar S
Sprint 1 Start / End	28-09-2026 / 28-09-2026
Sprint 2 Start / End	29-09-2026 / 30-09-2026
Sprint 3 Start / End	01-10-2026 / 02-10-2026
Team Roles	Abdul Wahith M — Backend & AI Integration; Kamalesh T — UI / Frontend Development; Bilu Besto H — Validation, Feedback & Reliability; Bharath Jeyakkumar S — Team Leader / Testing & Quality Assurance; Prasannavelan K — Database, Documentation & Demo Support
Maximum Marks	3 Marks

Solution Overview

FitBuddy is a web application that combines structured user validation, Google Gemini generation, persistence and feedback-based revision.

Core flow

1. User enters profile and fitness preferences.
2. Pydantic validates and normalizes the input.
3. Gemini generates a seven-day workout plan.
4. A separate Gemini Flash pathway generates a nutrition/recovery tip.
5. User and plan data are persisted in SQLite.
6. User can submit feedback.
7. The latest plan becomes the context for a revised plan.
8. The original plan remains stored.

Reliability behavior

- Configurable model names and fallback lists.
- Retry handling for transient AI errors.
- Static nutrition-tip fallback.
- Mock-AI mode for automated testing and demonstrations.

Future UI independence

The current Jinja2 presentation is not tightly coupled to the core business logic. A future React or other frontend can consume the JSON API while retaining the same generation and persistence behavior.